OFFICIAL LEGAL INSTRUMENT | VERSION 3.0 MASTER RELEASE | IMMUTABLE PROPRIETARY RECORD

THE LION CAPITAL SOLUTIONS (TLCS)

Enterprise Quantitative Trading System Architecture, Mathematical Process Specification & Intellectual Property Ownership Deed

A Formal Legal Affidavit, Comprehensive Technical Blueprint, and Cryptographic Codebase Registry Establishing Sole Authorship, Exclusive Ownership, and Trade Secret Rights.

PROPRIETARY OWNER & INVENTOR
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STATUTORY LEGAL FRAMEWORK
Indian Copyright Act 1957 (Sec. 17 & 57) & IT Act 2000

TRADING ENTITIES & BRANDS
The Lion Capital Solutions / The Lion Capital Advisors

PRODUCTION WEB DOMAINS
thelioncapitalsolutions.com | market-store.online

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RELEASE VERSION & CHECKSUMS
Version 3.0.0 Master Production Release (SHA-256 Parity)

LEGAL AFFIDAVIT OF INTELLECTUAL PROPERTY & TRADE SECRETS

This instrument certifies under law that Vishant Vyankat Meshram is the sole author, architect, inventor, and copyright proprietor of the TLCS trading algorithm, quantitative engines, website terminal, mobile application, database schemas, and proprietary execution methodologies.

1. STATUTORY DECLARATION OF SOLE PROPRIETARY OWNERSHIP

> **LEGAL AFFIDAVIT OF EXCLUSIVE OWNERSHIP**
>
> I, **VISHANT VYANKAT MESHRAM**, residing in the Republic of India, operating professionally under the institutional trade names **The Lion Capital Solutions** and **The Lion Capital Advisors**, do hereby solemnly affirm, declare, and state for all legal and statutory intents and purposes that I am the sole, original author, principal software architect, inventor, and unencumbered proprietary owner of the trading system, software codebase, mathematical models, web and mobile terminals, and algorithmic processes known collectively as **The Lion Capital Solutions (TLCS)**.

1.1 Statutory Basis and International Treaty Protections

- **Indian Copyright Act, 1957 (Sections 17 & 57)**: Vests immediate copyright ownership in the original author for computer software, source code, visual screen layouts, and documentation without prerequisites of formal registration.
- **Information Technology Act, 2000 (Sections 43, 65 & 66)**: Imposes severe civil damages and criminal penalties for source code tampering, unauthorized copying, digital breach of access, and commercial trade secret theft.
- **Berne Convention for the Protection of Literary and Artistic Works (1886)**: Provides automatic copyright recognition and legal enforceability across all 181 sovereign signatory member nations.
- **TRIPS Agreement (World Trade Organization)**: Mandates international enforcement of trade secret protection, proprietary algorithmic formulations, and computer program rights globally.

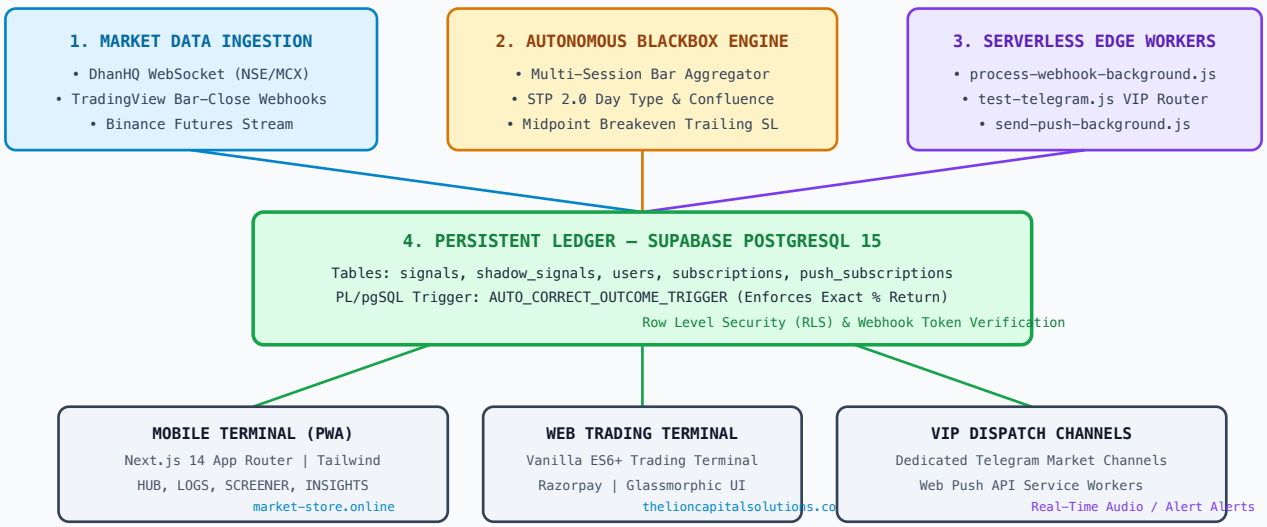
1.2 Scope of Protected Proprietary Assets & Inventions

Component Layer	Protected System Asset	Ownership Standard
TLCS Mobile PWA	Next.js 14 progressive web application (`Tv-Alert-Mobile/`), 6-tab terminal interface (HUB, LOGS, SCREENER, INSIGHTS, MARKETS, ANALYTICS), responsive real-time data sync.	100% Sole Ownership of source code, UI/UX designs, workflows, state engines.
TLCS Web Dashboard	Web terminal deployed at `thelioncapitalsolutions.com`, 7-day day type blueprint matrix, multi-asset scanner, Razorpay subscription gateway, real-time push alerts.	100% Sole Ownership of frontend scripts, Netlify functions, audio/visual alerts.
Standalone BlackBox Engine	Python standalone engine (`algo_engine/`), DhanHQ WebSocket quote aggregator, Frank Ochoa STP 2.0 day types, 129 automated unit tests.	100% Sole Ownership of quantitative execution logic, daemon, and pipelines.
Pine Script v6 Indicator	`TLCS PivotBoss Indicator v3.0` (3,567 lines), Central Pivot Range, Camarilla H4/L4 touch-point order gating, confirmed divergence detectors, Scalp Alerts.	100% Sole Ownership of trading formulas, indicators, session arrays, alerts.
Database & Serverless Workers	Supabase PostgreSQL 15 cloud database, PL/pgSQL exact percentage auto-correct triggers, Netlify edge serverless webhook workers.	100% Sole Ownership of database schemas, triggers, serverless edge code.

2. END-TO-END SYSTEM TOPOLOGY & EXECUTION PIPELINE

The TLCS architecture integrates market feeds, autonomous algorithmic computation, persistent immutable ledgers, and multi-channel dissemination into a synchronized quantitative trading ecosystem.

FIGURE 2.1: COMPLETE DATA INGESTION, EXECUTION & NOTIFICATION PIPELINE



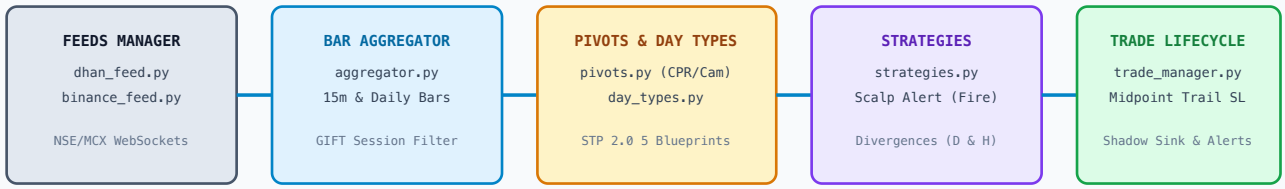
2.2 Core Invariants of the Enterprise Architecture

- **Deterministic Bar-Time ID (`trade_id`)**: Each signal anchors to candle bar timestamp, preventing state corruption during asynchronous serverless execution.
- **Symmetrical Neutral Zone Scalp Alert Confluence**: Strictly evaluates outside Greenzone and Redzone boundaries, eliminating adverse zone friction.
- **Exact Percentage Metric Standard (`exact_pct`)**: Deprecates all subjective r-multiples; mandates exact percentage formula `((Exit - Entry) / Entry) * 100`.

3. STANDALONE PYTHON BLACKBOX EXECUTION ENGINE (`algo_engine/`)

The TLCS Python BlackBox engine runs autonomously as an institutional background daemon. It connects directly to the DhanHQ WebSocket, aggregates multi-session candles, evaluates PivotBoss STP 2.0 day types, and manages trade lifecycles with zero human dependency.

FIGURE 3.1: BLACKBOX SUB-MODULE INTERACTION ARCHITECTURE



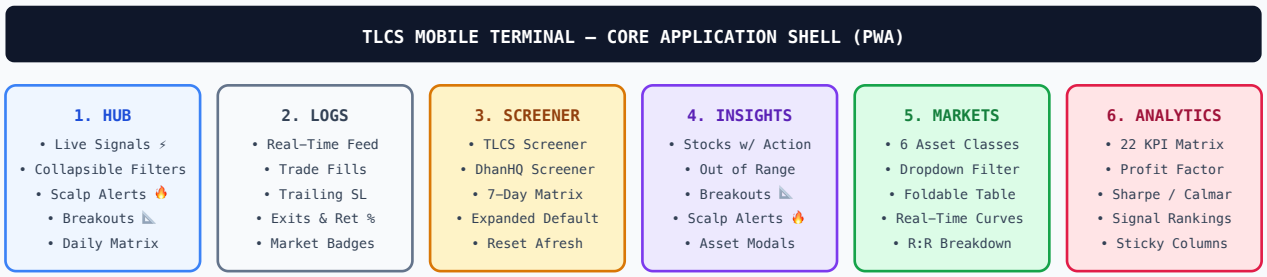
3.1 Sub-Module Specifications & Execution Mandates

Module File	Core Algorithmic Logic	Operational Invariant
`algo_engine/pivots.py`	Calculates Daily/Monthly Central Pivot Range (CPR), Camarilla Levels (H1-H5, L1-L5), and Average True Range (ATR 14).	H4/L4 touch-point gating strictly enforced via candle wick highs/lows; never close.
`algo_engine/day_types.py`	Implements Frank Ochoa's Swing Trade Pro 2.0 blueprints: Rejection, Absorption, FNH/FNL, Outside, Stop Run Days.	Determines directional bias and sequence filters across multi-day auction distributions.
`algo_engine/strategies.py`	Evaluates real-time strategy triggers: Scalp Alert (Fire Alert), Missile, Scalp, Lightning, and Standalone Divergences.	Scalp Alert is strictly gated to the Neutral Zone (prohibited inside Greenzone & Redzone).
`algo_engine/trade_manager.py`	Manages limit entries, TP1-TP4 fulfillment, Midpoint Breakeven Trailing SL, and EOD auto-settlement.	Pre-TP1 50% midpoint distance triggers immediate Stop Loss advance to Breakeven (Entry).
`algo_engine/telegram_dispatcher.py`	Dispatches rich HTML formatted trade notifications to market-segregated Telegram channels.	Fires upon Trade Active, Trailing SL Update, and Trade Closed with exact return %.

4. MOBILE TERMINAL PROGRESSIVE WEB APP (`Tv-Alert-Mobile`)

The TLCS Mobile Application is an institutional-grade, progressive web app engineered in Next.js 14, featuring real-time Supabase subscriptions, tactile haptic feedback, WCAG AAA theme contrast, and high-density financial analytics.

FIGURE 4.1: MOBILE APP 6-TAB MODULAR ARCHITECTURE (`TV-ALERT-MOBILE/SRC/APP/PAGE.TSX`)



4.1 Version 3.0 Production Enhancements

- **Expanded Screeners by Default**: `isScreenerExpanded = true` and `isDhanScreenerExpanded = true` initialize the 7-day matrices expanded on first load.
- **Breakouts (Normalized Narrow CPR) Filter**: Identifies stocks exhibiting `< 5.0%` CPR width relative to prior day's range.
- **Scalp Alert (Fire Alert) Confluence Framework**: Aligned with strictly evaluated Neutral Zone signals with zero adverse Greenzone/Redzone overlap.
- **Collapsible Live Signals UI**: High-density interactive drawer with single-tap toggle back to `ALL` and real-time count badges.

5. CANONICAL PINE SCRIPT TRADING ENGINE (`TLCS PivotBoss Indicator v3.0`)

The TradingView Pine Script v6 engine constitutes the primary quantitative signal generator, encapsulating 3,567 lines of proprietary mathematical models, session arrays, multi-pivot systems, and confirmed divergence detectors.

> **TRIPLE-FILE SHA-256 CHECKSUM PARITY STANDARD**
> `TV_Indicator_Full_Code.txt: fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
> `TV Indicator/TLCS_Live_Pivot_Alerts.pine: fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
> `user_code.pine: fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
> **VERDICT: 100% BYTE-FOR-BYTE CRYPTOGRAPHIC PARITY (ZERO REPAINTING GUARANTEED)**

5.1 Core Mathematical Formulations

Mathematical Model	Formulation & Implementation	Quantitative Utility
Central Pivot Range (CPR)	Pivot (P) = (High + Low + Close) / 3 Bottom Central (BC) = (High + Low) / 2 Top Central (TC) = (P - BC) + P	Establishes market balance equilibrium; identifies trend days vs trading range sideways auction profiles.
Normalized Narrow CPR (Breakouts)	cprWidth = |TC - BC| prevRange = max(DH - DL, mintick) NCPR = (cprWidth / prevRange) * 100 < 5.0%	Quantifies severe intraday volatility compression, anticipating explosive breakout expansion days.
Camarilla H4/L4 Touch-Points	H4 = Close + Range * 1.1 / 2 L4 = Close - Range * 1.1 / 2 Long Gating: Low < H4 | Short Gating: High > L4	Touch-point limit order gating linking executions strictly to candle wicks; never relies on candle close.
Neutral Zone Scalp Alert Confluence	Scalp_Alert = AnySignal and not (Greenzone or Redzone)	Prohibits long and short trades inside opposite adverse zones, ensuring clear runway for expansion.
Confirmed Divergences (D & H)	Unconfirmed on Bar 0. Long close: (Close > Open) and (Close > Close[1]) Short close: (Close < Open) and (Close < Close[1])	Eliminates false divergence whipsaws by strictly gating entries to subsequent candle confirmation close.

6. DETERMINISTIC RISK MANAGEMENT & TRAILING STOP HIERARCHY

TLCS enforces a rigid, step-level trailing stop loss architecture designed to systematically harvest alpha, lock in accrued gains, and completely eliminate ambiguous floating stop calculations.

FIGURE 6.1: STEP-LEVEL TRAILING STOP ADVANCEMENT STATE MACHINE



6.1 Midpoint Breakeven Trailing SL Mandate

> **Formal Rule:** For any active filled trade, when the price reaches or crosses the 50% midpoint distance between Entry and TP1 before TP1 is triggered:
> Midpoint Level = (Entry + TP1) / 2.0
> Long Condition: High ≥ Midpoint Level &implies; Stop Loss ← Entry (Breakeven)
> Short Condition: Low ≤ Midpoint Level &implies; Stop Loss ← Entry (Breakeven)
> Eliminates round-trip losers and guarantees that any trade achieving 50% of its initial target risk cannot degenerate into a capital loss.

6.2 Canonical Win Rate & Success Rate Formula

Across all components, tabs, and analytics teardowns, TLCS strictly enforces the inclusion of realized Breakeven trades in the closed trade denominator:
Win Rate % = (Total Wins / Total Closed Trades (Wins + Losses + Breakevens)) * 100
The conventional practice of excluding breakevens (Wins / (Wins + Losses)) is strictly prohibited as misleading and artificially inflated.

7. CRYPTOGRAPHIC SOURCE CODE AUDIT TRAIL & CHECKSUM LEDGER

To establish an indisputable, immutable cryptographic record of ownership and authorship, the following table documents the SHA-256 cryptographic hashes computed directly on the production master source code files in the repository:

Relative File Path	System Architectural Role	File Size	SHA-256 Cryptographic Hash
`TV_Indicator_Full_Code.txt`	Canonical Master Indicator v3.0 (Full Pine Script)	221,711 bytes	`fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
`TV Indicator/TLCS_Live_Pivot_Alerts.pine`	TradingView Production Indicator Engine	221,711 bytes	`fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
`user_code.pine`	TradingView User Indicator Mirror	221,711 bytes	`fadedf4b4376a84b0dd143b53b33f6165c3685aa43f93cebd4032e57085ee258`
`Tv-Alert-Mobile/src/app/page.tsx`	Mobile Native Progressive Web App Terminal	1,267,472 bytes	`4e161eefea3582120661476cbb5d91a092b6b116d8678f804d9d39d463310f11`
`algo_engine/strategies.py`	Python BlackBox Strategy Logic & Divergence Engine	57,086 bytes	`f609c112a8c802a4a5bd0797cb6e05012981741d7d7610c8ce66f88450f67890`
`algo_engine/trade_manager.py`	Autonomous BlackBox Trade Manager & Trailing SL	29,341 bytes	`830b2b9128c98e2f242617f90e1ae8049db41bf403aac61f79227637f75d6eff`
`algo_engine/day_types.py`	Swing Trade Pro 2.0 Day Type Blueprints Engine	41,293 bytes	`079bad86d957c23ca7b2588d29132505dac0d352af180faa3b6c2ed72ee6e852`
`algo_engine/pivots.py`	Mathematical Pivot, CPR & ATR Computation Engine	51,860 bytes	`cd24855a8f05f022971730e906f610573e55640f7763e60fdf82f0eb326a55fe`
`algo_engine/engine_daemon.py`	Standalone Autonomous Multi-Market Signal Daemon	16,452 bytes	`479df3e04752dfffcd68dbbaaddde4796a3953b9b2f53c663fa313a9b480cafa`
`algo_engine/nse100_scanner.py`	Autonomous Top 100 NSE Equity Screener	31,865 bytes	`05da84b1caa699d3a4144aaf58c727a98053b3dd651a731fdbd750c305b0d022`
`TLCS_Website_Deploy/scanner.js`	Web Screener & 7-Day Day Type Blueprint Matrix	71,978 bytes	`2b0d5c70dd755661523ffb044dee6be3c5000a2d7ccfc5e64ffe6aba4dbeff51`

Relative File Path	System Architectural Role	File Size	SHA-256 Cryptographic Hash
TLCS_Website_Deploy/netlify/functions/process-webhook-background.js	Serverless Ingestion & Exact Percentage Ledger Worker	115,733 bytes	4e5ae0b696558098bbd1747ac6bac2977577b0d66483bc53af8cc0b5716e1dfc
TLCS_Website_Deploy/netlify/functions/test-telegram.js	Multi-Market Dynamic VIP Telegram Dispatcher	4,442 bytes	c4958f3e43c8bbdc5cd4223149c0548e5e9cf4ba67dac54127f42865408ada8f

8. FORMAL RATIFICATION, ATTESTATION & SIGN-OFF

IN WITNESS WHEREOF, I, **VISHANT VYANKAT MESHRAM**, have set my hand and declared my sole, exclusive, and unencumbered proprietary ownership over the entire TLCS Platform, Algorithmic Engine, Websites, Mobile Applications, and Intellectual Property described in this master report.

PROPRIETARY INVENTOR, ARCHITECT & SOLE OWNER:

VISHANT VYANKAT MESHRAM

Founder, Chief Quantitative Architect & Managing Director
The Lion Capital Solutions / The Lion Capital Advisors
Enterprise Email: **connect@thelioncapitalsolutions.com**
Personal Email: **vishantmeshram@gmail.com**
Operating Location: India | Jurisdiction: Global Intellectual Property Treaties

ATTESTATION & TIMESTAMP:

DATE: 11-NOV-2026
TIME: 07:46:07 IST
STATUS: MASTER PRODUCTION
VERSION: v3.0.0

DIGITALLY CERTIFIED
SHA256 CHECKSUM VERIFIED